

Bhopal Problem Statements

Four problem statements for a one-day civic build in Bhopal, Madhya Pradesh, written for partner organisations, mentors and judges. One statement is selected with the partner for the build day.

FORMAT

One-day build, teams of four to five, six to eight hours

OWNERSHIP OF RESULTS

Open source, donated to the partner

Summary

Each statement names one primary user, one workflow and one measurable outcome, and is scoped to be buildable by a team of four or five in a single day using Claude.

ID	TITLE	THEME	PRIMARY USER	SOLUTION TYPE
PS-1	Is this a scam?	Financial safety and consumer protection	An individual who has just received a suspicious message, call or app prompt	Conversational assistant with image and text understanding over a curated reference layer
PS-2	What should I learn next?	Skills and digital literacy in a Tier-2 city	A student or working learner acquiring technology skills outside a formal institution	Guided learning-pathway generator over a curated resource catalogue
PS-4	From a voice note to a job-ready profile	Employment and career guidance	A first-generation job seeker, typically 12th pass or holding a trade certificate	Speech-to-profile pipeline with eligibility-aware matching and document generation
PS-5	Triage the city's complaint queue	Civic service delivery and departmental accountability	The complaint desk operator at a municipal zone office	Intake classification and routing engine with reporting

profile on the key fields, and shortlists match the counsellor's selection in the majority of cases.

- A job seeker completes the process in one sitting and leaves with a resume, a shortlist and interview preparation.
- The partner institution confirms it would use the tool at its next placement drive.

PS-5

Triage the city's complaint queue

THEME	PRIMARY USER	SECONDARY USERS
Civic service delivery and departmental accountability	The complaint desk operator at a municipal zone office	Zone heads, department leads, field teams

Background

Residents of Bhopal report civic issues through several parallel channels: the state CM Helpline (181), the Municipal Corporation's helpline and mobile app, direct messages to elected representatives, and social media. Complaints arrive as text, voice notes and photographs. They mix languages, describe locations informally, and are often filed by several residents for the same incident through different channels, with the responsible department guessed by the complainant. Zone office staff read, classify, route and follow up by hand. Departmental performance is not compared systematically.

Problem statement

The city's complaint intake is unstructured, duplicated and manually routed, which delays resolution, obscures accountability and consumes staff time on clerical work rather than follow-up.

A citizen-facing portal is out of scope for a one-day build. The operator-facing workflow inside it is not.

Proposed solution

An engine that accepts raw complaints, as text, voice or photograph with caption, from an exported dataset and produces structured tickets with department, category, urgency, normalised ward or locality, duplicate linkage and language. It drafts an acknowledgement for operator approval and generates a weekly per-department digest: complaints received, resolved, median resolution time, and

repeat complaints by locality. The engine is read-only over the export and does not connect to any live system.

Key capabilities

1. **Multi-modal intake.** Text, speech transcription, and image with caption.
 2. **Classification.** Assignment to the partner's department and category taxonomy, with a confidence level and the terms that determined the routing.
 3. **Urgency scoring.** Against agreed criteria: public safety, service outage, duration of the problem.
 4. **Locality normalisation.** Mapping to ward using a gazetteer that includes colloquial place names.
 5. **Duplicate and cluster detection.** Across channels and time windows.
 6. **Acknowledgement drafting.** For operator approval before anything is sent.
 7. **Reporting.** A weekly digest per department, with emerging-cluster alerts as an extension.
-

Inputs and data sources

- A labelled dataset of complaints with department, category and urgency: either an anonymised export from the partner or a synthetic corpus reviewed by a partner official for realism.
 - The partner's department and category list.
 - A ward and locality gazetteer with aliases.
 - Public department-level summaries from the state helpline, for the digest format.
 - A held-out test set with partner-agreed labels.
-

Constraints

- No write access to the helpline, the municipal system or any live channel. Export in, tables and reports out.
 - Anonymised data only: names, phone numbers and house numbers removed, dates shifted, small categories suppressed.
 - Human in the loop. The operator approves every acknowledgement; the engine routes, a person confirms.
 - Routing is explainable at the level of the individual ticket.
-

Expected outcomes and success criteria

- Routing accuracy per department on the held-out test set, reported by the team.
- Duplicate detection reduces the ticket count on the test set in line with the

partner's expectation.

- The weekly digest is judged usable by a zone officer without modification.
 - The partner confirms it would run the engine on the previous week's export.
-

Public references: the National Cybercrime Reporting Portal and 1930 helpline; the Reserve Bank of India's published list of digital lending apps of regulated entities; the National Career Service and national apprenticeship portals; the Madhya Pradesh CM Helpline. Identifiers follow the original numbering; PS-3 was withdrawn.